



Model Invariance

Runtime Qualification

Frozen NAO ROS4HRI E2E model comparison

Prepared for TFM review
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Executive summary

The frozen NAO ROS4HRI stack passed the required startup gate with three Ollama models. The requested vLLM endpoint was unavailable during the final inventory probe, so it is recorded as a blocked semantic cell.

The two Gemma variants reproduced the same main and extreme-capability failure shape while passing the environment, KB, and deterministic fake-skill success families. Nemotron retained environment execution and targeted fail-once recovery but introduced additional planner coverage, KB wording, trace-correlation, and report-closure variance.

At a glance

- Three model cells
- Gemma4 variants shared the failure intersection; this is the primary evidence against attributing every failure to model quality.
- No vLLM semantic result was scored because /v1/models returned HTTP 000.

Introduction

The comparison keeps the frozen image, ROS graph, prompt and generation settings, KnowledgeCore fixtures, fake-skill policies, questionnaire cases, and speech evidence policy constant. The only experimental factor is the selected Ollama model, with the declared no-driver profile used for the two alternative cells after the external NAOqi endpoint timed out.

Live questionnaire rows are runtime stress evidence over fixed cases. They are distinct from deterministic source and harness tests. A pass proves the expected evidence path for that case, not universal success over all natural-language inputs.

Key findings

The strongest invariant is the 11/11 environment result for all three models. The Gemma variants also passed 7/7 KB stress and 9/9 all-success fake-deep cases. Nemotron passed 4/6 scoreable KB cases and 8/9 fake-deep cases, with its targeted fail-once navigation probe passing with observed failure and replan.

Context and conditions

The standard-family strict pass rates were 84.7% for gemma4:31b-cloud, 85.0% for gemma4:cloud, and 76.8% for nemotron-3-super:cloud. These are one-run engineering rates, not a universal model ranking.

Patterns in the evidence

Shared failures concentrated in generic pick admission, maximal kitchen delivery, missing-recipient clarification, universal target selection, and high-composition capability coverage. Nemotron-only variance concentrated in planner coverage, KB speech projection, trace correlation, and terminal report closure.

Key takeaway. The stack contract remained usable across the tested models, while generated route, target, plan, wording, and closure behavior remained model-dependent.

Implications

A model replacement must be evaluated through the same frozen E2E matrix. Endpoint reachability and one successful dialogue establish operational availability, not semantic equivalence.

Theme	Observation	Implication
Endpoint	vLLM /v1/models unavailable	No vLLM semantic score was fabricated
Cells	Gemma4 31B, Nemotron 3 Super, Gemma4 cloud	Three Ollama models passed startup preflight
Evidence	191 retained case rows	Raw JSON and derived thesis dataset preserved

Recommendations

Recommendations

1. Keep vLLM as the preferred backend only after /v1/models and a named completion probe succeed.
2. Use an explicit, launch-time fallback order and pin the selected model for the run. Record fallback reason and model identity in JSONL.
3. Repeat shared failures and model-specific sensitive cases at least three times before changing stack code or prompt policy.

Conclusion

The comparison supports a model-agnostic stack claim at the ownership and execution boundaries, with a qualification that model-mediated planning remains variable.

The retained raw JSON, dataset, hypothesis registry, and HTML report make the attribution auditable for thesis Results and Discussion writing.

Appendix

Notes

Evidence semantics

Deterministic tests validate implementation contracts. Live cases exercise endpoint reachability, ROS lifecycle, grounding, planning, execution, speech, and trace seams. The categories must remain separate in the thesis.

Source placeholders

- NAO ROS4HRI runtime stack. Model invariance E2E comparison and raw JSON artifacts. Local project evidence pack, 21 July 2026.
- Robot runtime performance review. REPORT.md and REPORT.html for the frozen v34 image. Local project artifact, 21 July 2026.
- Model comparison dataset. dataset.csv, dataset.jsonl, and comparison_summary.json. Generated from the retained live questionnaires, 21 July 2026.
- Backend inventory. vLLM and Ollama endpoint probe with selected model identities. Local runtime evidence, 21 July 2026.

Template note. The retained design-report visual system was used. The generated DOCX was rendered and checked after generation; raw evidence remains alongside this report.

Comparative evidence tables

The following tables summarize the JSON outputs. Source-file paths and full case observations are retained in the accompanying dataset.

Case family	Gemma4 31B	Gemma4 cloud	Nemotron 3 Super
Environment	11/11	11/11	11/11
Main	17/20	18/21	17/19 + 1 degraded
KB stress	7/7	7/7	4/6
Robustness	3/5	3/5	3/5
Fake deep all success	9/9	9/9	8/9 + 1 degraded
Capability extreme	3/7	3/7	0/6 + 1 not scored
Fail-once navigation	diagnostic full run	1/1	1/1

Startup and endpoint conditions

Condition	Observation	Interpretation
vLLM inventory	HTTP 000 at 10.7.138.215:8004	Blocked semantic cell
Ollama inventory	11 advertised models	Candidate source available
Chatbot and planner preflight	Pass for all three selected models	Launch readiness established
Required lifecycle nodes	Active [3] in each scored cell	Core graph ready
NAOqi endpoint	Connection timeout	Isolated from alternative semantic cells

Shared and model-specific outcomes

Evidence	Finding	Attribution posture
Gemma intersection	Same generic pick, maximal delivery, missing-recipient, and extreme coverage failures	Investigate stack or shared contract seams
Nemotron variance	KB wording omissions, partial report closure, planner coverage, and trace inconsistency	Model/backend or observability candidates
Fake recovery	Targeted fail-once navigation passed for Nemotron and Gemma cloud	Executor recovery seam remains viable
Environment	11/11 for all models	No broad grounding outage supported

Seam hypothesis adjudication

Hypothesis	Evidence	Status
H-01 runtime wiring	All Ollama preflights passed; vLLM unavailable	Ollama route ruled out; vLLM blocked
H-02 model policy	Gemma intersection stable; Nemotron adds variance	Supported
H-03 grounding and KB	Environment stable; Nemotron speech terms varied	Broad outage rejected
H-04 executor and recovery	Fail-once navigation passed with replan	Core supported; closure open
H-05 observability	Trace inconsistency and global timeouts retained	Active limitation
H-06 external service	vLLM unavailable; Ollama preflight stable	Backend condition confirmed

The shared Gemma failure shape is the key adversarial result. It prevents a simple model-quality explanation and directs the next investigation toward planner admission, target normalization, clarification speech, and high-composition plan coverage. The Nemotron-only cases remain candidate model variance until the same prompts are repeated with stable trace and speech correlation.

Suggested TFM wording

The evaluation shows that the ROS4HRI stack preserves its principal ownership and execution contracts across the tested language-model backends. Generated target selection, multi-step plan coverage, clarification wording, and terminal report closure vary with the selected model. Shared failures across the two Gemma variants were retained as stack-seam hypotheses rather than attributed to model quality alone.

Reproducibility and handoff

- Use analysis/comparison_summary.json for model-level tables and shared-failure intersections.
- Use analysis/dataset.csv for spreadsheet analysis and analysis/dataset.jsonl for programmatic TFM processing.
- Use raw per-model JSON when quoting exact speech, route, target, lineage, or timing evidence.
- Do not merge deterministic source-test counts into the live runtime stress denominator.
- When vLLM returns a stable model identity, repeat the same case matrix before promoting it as a replacement.

Appendix: provenance

The evidence pack preserves the endpoint probe, frozen image identity, startup snapshots, per-model questionnaire outputs, derived tables, methodology, and seam audit. The comparison is intentionally bounded to one frozen run per model. Repeated sensitive cells are the next acceptance gate for a model replacement.

Artifact	Purpose
REPORT.md and REPORT.html	Readable runtime review and thesis-facing discussion
analysis/comparison_summary.json	Model, suite, startup, and shared-failure summary
analysis/dataset.csv	Tabular per-case dataset with provenance
analysis/dataset.jsonl	Machine-readable per-case dataset
raw/	Authoritative JSON outputs for all model cells
../model_invariance_seam_audit_2026-07-21.md	Hypothesis registry and adversarial attribution